

Exam Project

EXAM FORMAT

The exam for *Introduction to Web Applications* consists of two closely connected parts, both of which are mandatory:

1. Individual design and development of a web application
2. Demonstration and oral discussion of the project.

The evaluation of the projects and the demonstration/oral discussion will concern the material submitted before the exam session. The demonstration and brief oral discussion regarding the project must be carried out according to the schedule that will be made available a few days before the exam session, after registration for the exam session itself.

The discussion will focus on the design choices, such as layout, code structure, database structure, etc., as well as the implementation and functional choices adopted.

DESCRIPTION AND REQUIREMENTS

The project consists of a web application that must satisfy a number of technical, stylistic, and functional requirements, detailed below.

The web application must use the technologies illustrated and tested during the course.

Logistical Requirements

- The project must be carried out individually.
- A submission that is partially or entirely shared with another student enrolled in the course is neither allowed nor acceptable.
- The project must be submitted according to the deadlines indicated on the last page of this document.
- Late submissions are neither allowed nor acceptable.

Technical requirements

The web application must comply with the following technical requirements:

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- Use of HTML5 and CSS3, relying, if necessary, on external frameworks such as Bootstrap, while customizing their style through ad hoc rules.
 - Use of Flask and SQLite, as a relational database, for the back end.
 - Use of Flask-Login for authentication management.

In addition:

- The web application must have a clearly defined target device type, chosen by the students: desktop, mobile, or tablet, optionally supporting responsive design.
- Any forms used in the project must be validated both on the front end and on the back end, as explained during the course.
- The submitted project must be fully testable by the instructor and must work on the latest versions of Chrome, 111+, and Firefox, 110+.
- The source code must be well written and include appropriate comments where necessary.
- All the technologies listed above must be integrated cohesively and consistently within a single web application.

Required:

- Deploy the web application on PythonAnywhere: <https://www.pythonanywhere.com/>

Stylistic Requirements

The web application must comply with the following stylistic requirements:

- Use HTML tags semantically, for example, not everything should be a <div>.
- Do not use deprecated HTML tags.
- Do not use inline CSS declarations; CSS rules must be kept separate from the semantic structure of the HTML.

In addition, the web application must be sufficiently usable.

Description: Functional requirements

The goal is to create a web application for managing a *fantasy adventure guild*. For simplicity, the application will be dedicated to a **fictional** single week of activities, from Monday to Sunday. **The days of the week are not linked to real calendar dates. The application will, therefore, use a simulated *current day and time* within the fictional week.**

The application must support two types of registered users: **adventurers** and the guild administrator, called the **Guild Master**.

Adventurers can explore the program of quests scheduled during the week and join quest sessions. To join a quest, an adventurer must be registered and logged in. Login and registration

require a unique field that will be used to identify the user on the website, such as an email address or a username.

The Guild Master, who must also be authenticated through login, can manage the weekly quest program. In particular, the Guild Master can create new quests and schedule the related quest sessions. When a new quest session is created, the system must prevent overlaps in the same day, time and location, so that each location hosts only one quest session at a time. Quest sessions can be freely modified as long as no adventurer has joined them, including date, starting time and location. A quest session can be cancelled only if no adventurer has joined it. Once created, a quest cannot be modified.

Each quest available on the website must have the following mandatory information associated with it:

- Quest title
- Duration, in minutes
- Quest type, for example combat, exploration, puzzle, stealth, magic or survival
- Difficulty level, for example easy, medium, hard or legendary
- Short description
- Promotional image or illustration
- List of quest sessions connected to the quest, including day, starting time and location

Each quest session will take place in one of three available locations, chosen by the students and consistently used throughout the application, such as Dungeon Hall, Enchanted Forest, Wizard Tower. Each quest session supports three categories of party roles:

- Warrior—4 available places per quest session
- Mage—3 available places per quest session
- Healer—2 available places per quest session

Places are not numbered. Each adventurer can reserve at most 2 places for the same quest session, for example to bring a companion. Both places must be associated with the same selected role category.

Each adventurer can join at most 3 quest sessions during the week. An adventurer cannot join two quest sessions that overlap in time. Once all places for a given role category have been taken, it will no longer be possible to join that quest session with that role.

Quest participations can be modified **only if the related quest session starts more than 8 hours after the simulated current day and time**. After this deadline, the participation can no longer be modified or cancelled.

On the homepage of the application, all users, registered or not, will see a short version of the available quest program, ordered by day and increasing starting time. Through specific filters, users will be able to explore quest sessions by day, quest type, difficulty level or available role.

By clicking on a quest session, users will be able to view all detailed information.

Once an adventurer has joined a quest session, the participation will be visible in the adventurer's profile page, where the quest title, day, starting time, location, selected role category and number of reserved places will be shown.

The Guild Master will also have a personal profile page showing all scheduled quest sessions, grouped or linked by their respective quests. For each session, the Guild Master can modify it if no adventurer has joined yet, or cancel it.

In addition, the Guild Master will be able to view participation statistics for each quest session, such as the number of reserved places, the remaining places for each role category and the most requested roles.

The Guild Master can browse the website like adventurers, but cannot join quest sessions. An adventurer cannot become a Guild Master. Unregistered users can freely explore the website, but they cannot create quests or join quest sessions.

Note: upon submission, the web application must contain some sample data allowing the instructors to test its main functionalities. In particular, it must include at least 5 registered users: 1 Guild Master and 4 adventurers. It must also contain at least 6 quests, with at least 2 quest sessions for each day. At least one quest session must still be modifiable because it has no participants. There must also be existing participations for all three role categories, with at least one Warrior, one Mage and one Healer already reserved. At least one role category in one quest session must be fully booked.

The application must allow the instructors to test both modifiable and non-modifiable participations according to the **simulated current day and time**. The usernames and passwords of the sample accounts, **together with the simulated current day and time**, must be provided to the instructors for verification.

🔥 Students are encouraged to fully express their creativity in the choice of the guild name, quest names, fantasy world, visual theme, characters, locations, and the overall experience offered to users. The goal is to create an original and personalized project.

OPTIONAL EXTRA REQUIREMENT FOR PROVA FINALE

Students who need to complete the Prova finale may extend the project by implementing an additional platform administrator profile, the **Guild Council administrator**. For simplicity, the application will include only one Guild Council administrator account.

The Guild Council administrator does not create quests and cannot join quest sessions. Instead, they have access to a dedicated administration page where they can view general information about the guild and its weekly activity.

In particular, the Guild Council administrator must be able to:

- View the list of all registered adventurers.

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- View the main information associated with each adventurer, such as username or email address and number of quest participations.
 - View the list of all quests and quest sessions scheduled during the week.
 - View the details of each quest session, including day, starting time, location, quest type, difficulty level, available role categories and current participations.
 - View basic statistics about the platform, such as:
 - total number of registered adventurers;
 - total number of quests;
 - total number of quest sessions;
 - total number of participations;
 - total number of reserved places per role category;
 - most popular quest type;
 - quest session with the highest number of reserved places.

This additional functionality is optional and **is required only for students completing the Prova finale**. It should be integrated into the same web application in a coherent way, without replacing or simplifying the mandatory requirements described above.

PROJECT SUBMISSION INSTRUCTIONS

The project must be submitted through the “**Consegna Elaborati**” tool available on the course page on the Teaching Portal, by **23:59 on 15 July 2026**.

The submission must contain a single archive file, **.zip**, including:

- Source code of the developed web application, including:
 - any dependencies, in a file such as **requirements.txt**;
 - images;
 - etc.;
 - the SQLite file containing the database.
- A text document, **.md** or **.txt**, containing:
 - the user credentials;
 - any instructions needed to test the web application;
 - the url where the deployed web application is available.